

Circuit Simulation Lab:24

Design of Counters

The purpose of this experiment is to introduce the design of Johnson Counter.

In Johnson counter, the complemented output of last flip flop is connected to input of first flip flop and to implement n-bit Johnson counter we require n Flip-flop

CODE:

```
module johnson_counter(  
input clk,  
input rst,  
output reg [3:0] q  
);
```

```
always @(posedge clk)  
begin
```

```
if(rst)  
    q <= 4'b0000;
```

```
else  
    q <= {q[2:0],~q[3]};
```

```
end
```

```
Endmodule
```

```
module tb_johnson_counter;
```

```
reg clk,rst;  
wire [3:0] q;
```

```
johnson_counter dut(  
    .clk(clk),  
    .rst(rst),  
    .q(q)  
);
```

```
always #10 clk = ~clk;
```

```
initial begin
```

```
$dumpfile("johnson_counter.vcd");  
$dumpvars(0,tb_johnson_counter);
```

```
clk=0;  
rst=1;
```

```
#20 rst=0;
```

```
#250;
```

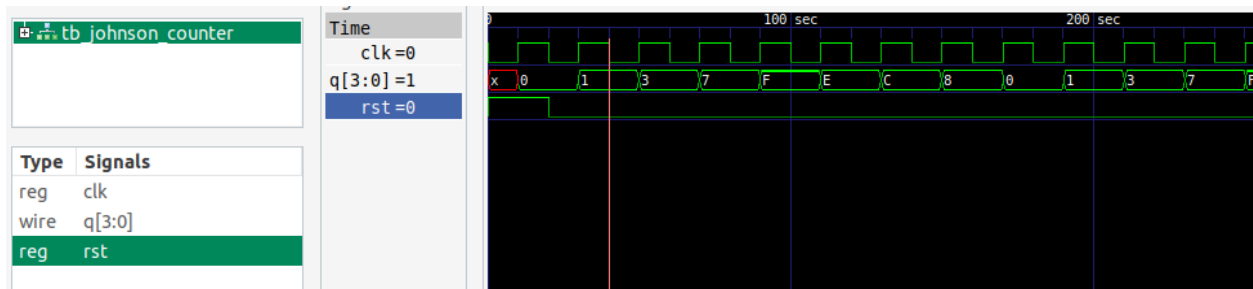
```
$finish;
```

```
end
```

```
initial
```

```
$monitor("time=%0t q=%b", $time, q);
```

```
Endmodule
```



```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_24$ iverilog -o out  
johnson_counter.v tb_johnson_counter.v  
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_24$ vvp out  
VCD info: dumpfile johnson_counter.vcd opened for output.  
time=0 q=xxxx  
time=10 q=0000  
time=30 q=0001  
time=50 q=0011  
time=70 q=0111  
time=90 q=1111  
time=110 q=1110  
time=130 q=1100  
time=150 q=1000  
time=170 q=0000
```